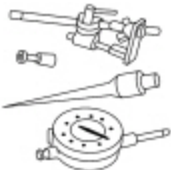
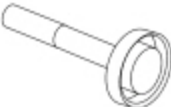


Differential Case and Ring Gear — Electronic Locking Differential (ELD)

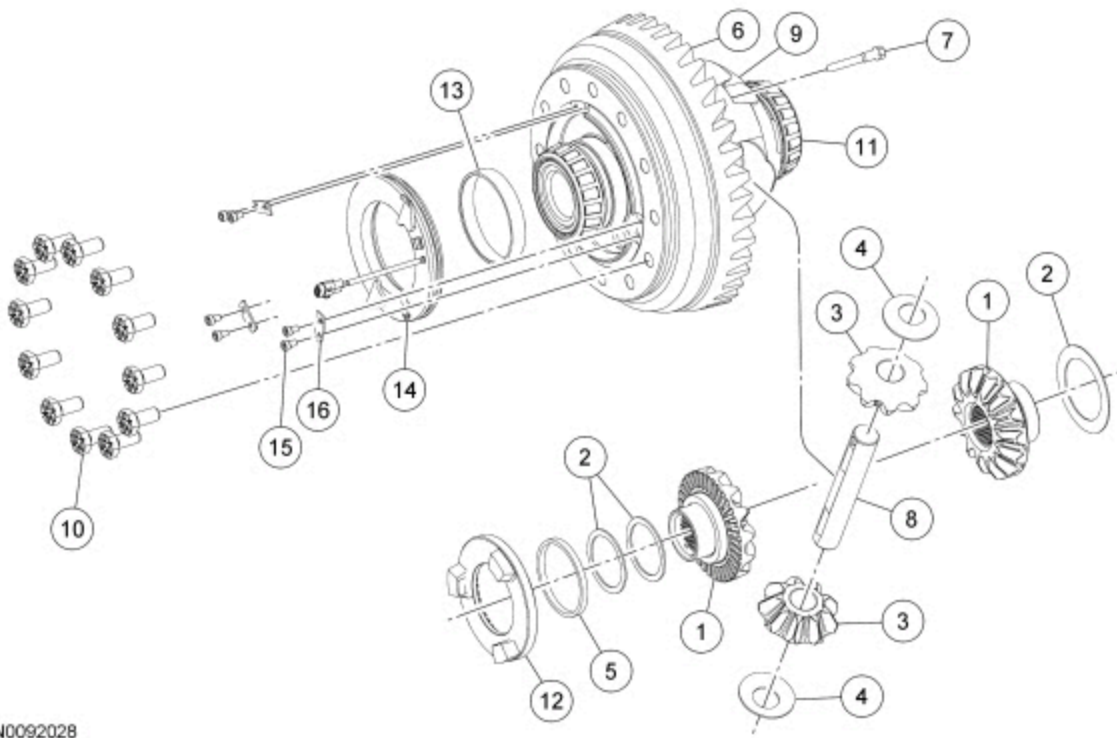
 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST2026-A	2 Jaw Puller 205-D072 (D97L-4221-A) or equivalent
 ST1214-A	Dial Indicator Gauge With Holding Fixture 100-002 (TOOL-4201-C)
 ST2473-A	Installer, Differential Carrier Bearing 205-D044 (D81T-4221-A) or equivalent
 ST1368-A	Puller, Bearing 205-D064 (D84L-1123-A)
 ST1543-A	Step Plate 205-D061 (D80L-630-8) or equivalent

Material

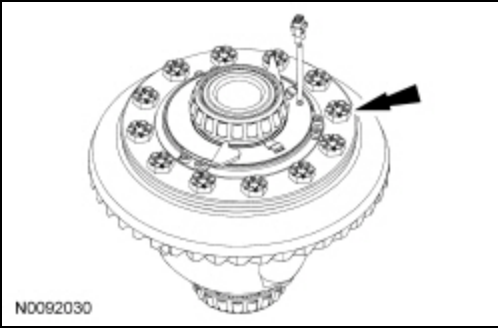
Item	Specification
Maximum Strength Retaining Compound Loctite® 638™	—
Motorcraft® SAE 75W-140 Synthetic Rear Axle Lubricant XY-75W140-QL (US); CXY-75W140-1L (Canada)	WSL-M2C192-A and GL-5



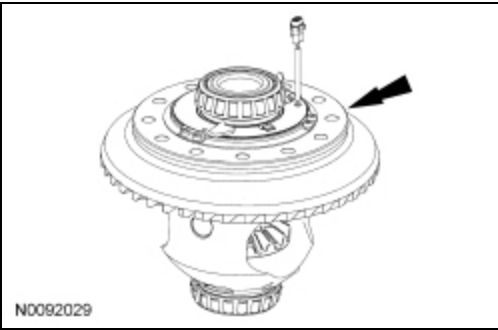
Item	Part Number	Description
1	—	Differential side gears (part of kit 4215)
2	—	Differential side gear thrust washers (part of kit 4215)
3	—	Differential pinion gears (part of kit 4215)
4	—	Differential pinion thrust washers (part of kit 4215)
5	—	Differential pinion spring (part of kit 4215)
6	4209 4209	Differential ring gear
7	4241 4241	Differential pinion shaft lock bolt
8	4211 4211	Differential pinion shaft
9	4204 4204	Differential case
10	4216 4216	Differential ring gear bolt (12 required)
11	4220 4220	Differential bearing
12	—	Cam-ring (part of kit 4880)
13	—	Plunger (part of kit 4880)
14	—	Coil (part of kit 4880)
15	—	Coil bolt (part of kit 4880) (6 required)
16	—	Stopper (part of kit 4880) (3 required)

Disassembly

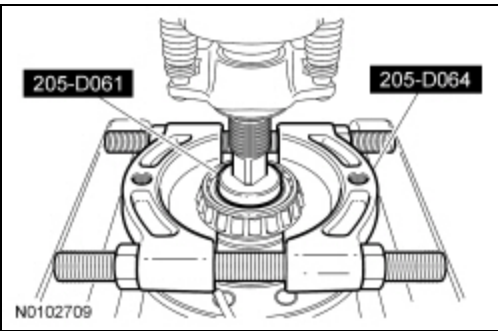
1. Remove the differential carrier. For additional information, refer to Differential Carrier in this section.
2. Remove and discard the 12 differential ring gear bolts.



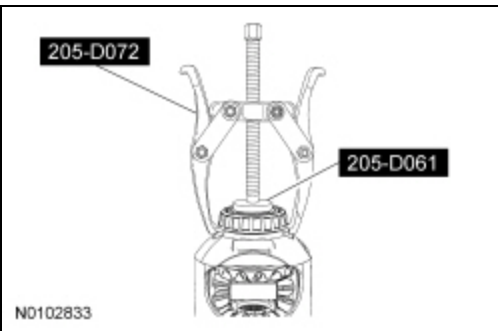
3. Insert a punch in the differential ring gear bolt holes and drive the differential ring gear off.



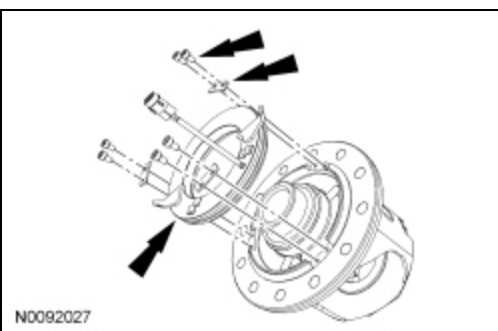
4. Using the Bearing Puller plate, Step Plate and a suitable press, remove the LH differential bearing.



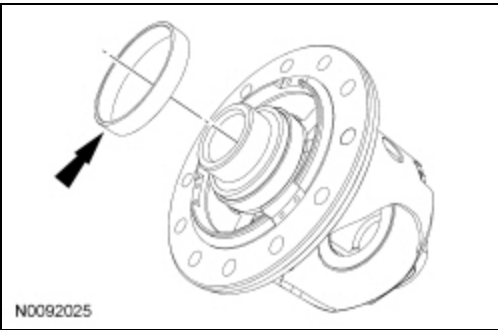
5. Using the 2 Jaw Puller and Step Plate, remove the RH differential bearing.



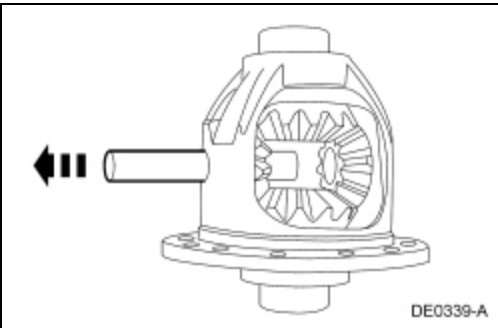
6. Remove and discard the 6 screws and 3 stoppers that hold the Electronic Locking Differential (ELD) coil in place and remove the coil.



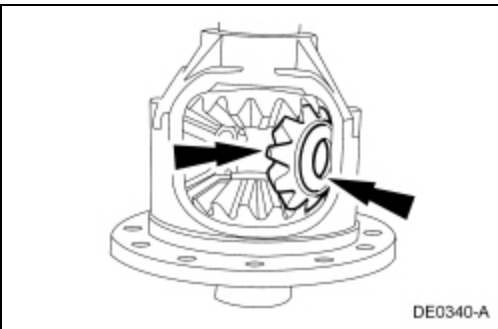
7. Remove the plunger assembly.



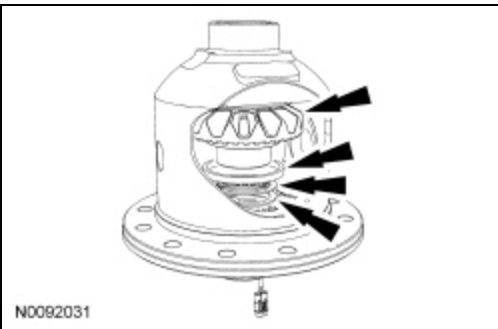
8. Remove the differential pinion shaft lock bolt and the differential pinion shaft.



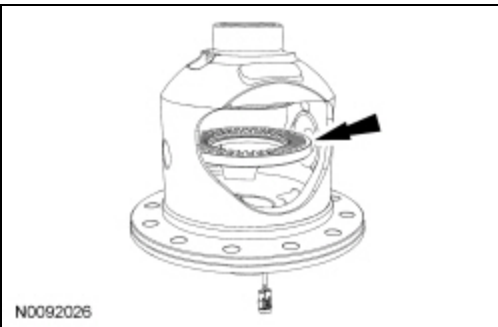
9. Rotate and remove the differential pinion gears and differential pinion gear thrust washers.



10. Remove the differential side gears, the differential side gear thrust washers and spring.



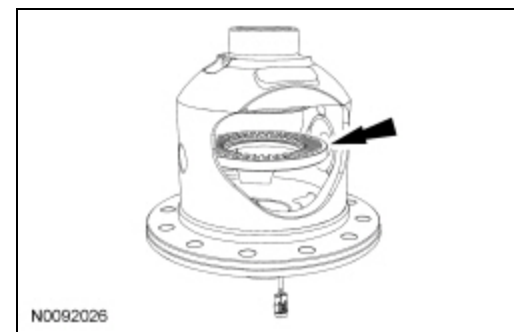
11. Remove the cam-ring.



Assembly

All vehicles

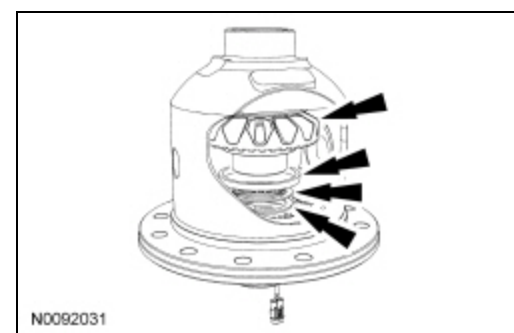
1. Install the cam-ring.



2. **NOTE:** Start by using the thickest thrust washers. If the gears do not assemble using the thickest thrust washers, back down in thickness until the gears will assemble.

Position the 2 LH differential side gear thrust washers and spring on the lower differential side gear and install the gear.

- Lubricate the differential side gear thrust washers and the differential side gear journals with axle lubricant.



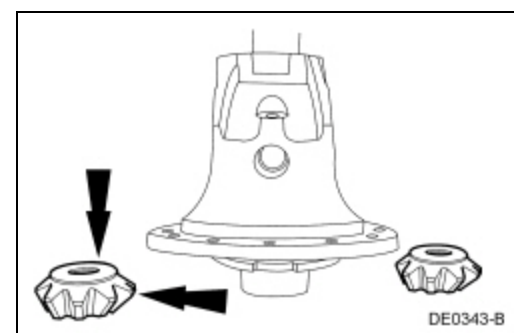
3. **NOTE:** Start by using the thickest thrust washer. If the gears do not assemble using the thickest thrust washers, back down in thickness until the gears will assemble.

Position the differential side gear thickest thrust washer on the RH differential side gear and install the gear.

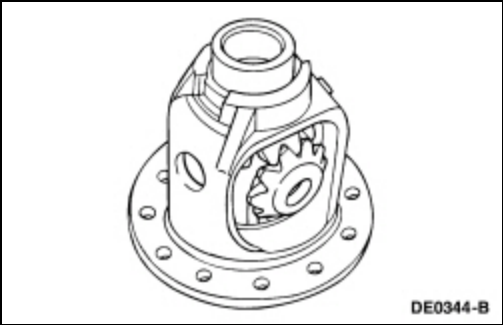
- Lubricate the differential side gear thrust washers and the differential side gear journals with axle lubricant.

4. Assemble the differential pinion gear thrust washers and the differential pinion gears.

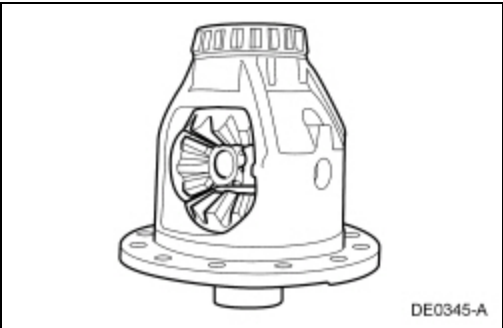
- Lubricate the washers and gears with rear axle lubricant.



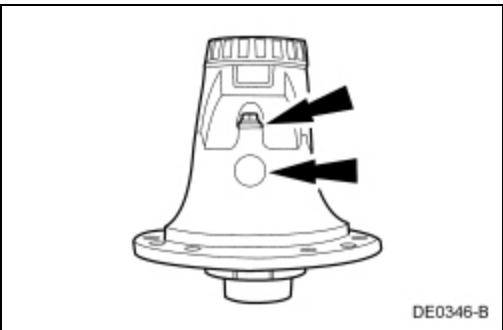
5. Engage the differential pinion gears opposite the differential side gears.



6. Rotate the differential pinion gears to align the differential pinion shaft bore.

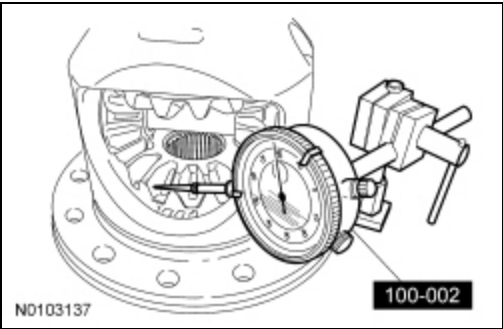


7. Insert the differential pinion shaft, and install a new differential pinion shaft lock bolt finger-tight.



8. Install the Dial Indicator Gauge With Holding Fixture.
- While pulling up on the RH differential gear, measure the backlash of the LH differential gear.
 - Backlash should be 0.001-1.000 mm (0.004-0.040 in).
 - If backlash is incorrect, install a thicker LH thrust washer to reduce backlash, or a thinner washer to increase backlash.

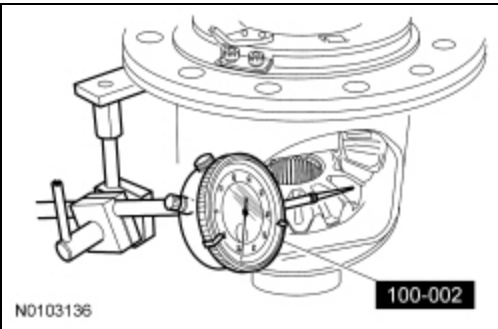
Part Number	Thickness (mm)	Thickness (in)	Quantity
Part of kit 4880	1.4	0.05	2
Part of kit 4880	1.6	0.06	2
Part of kit 4880	1.8	0.07	2



9. Turn the differential over and reposition the Dial Indicator Gauge With Holding Fixture.

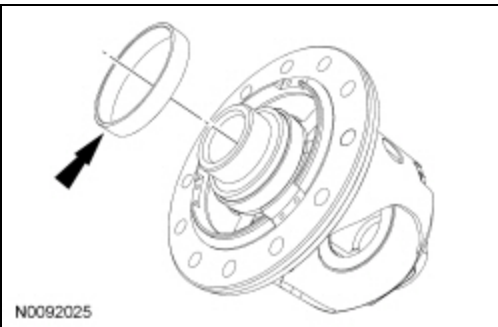
- While pulling up on the LH differential gear, measure the backlash of the RH differential gear.
- Backlash should be 0.001-1.000 mm (0.004-0.040 in).
- If backlash is incorrect, install a thicker RH thrust washer to reduce backlash, or a thinner washer to increase backlash.

Part Number	Thickness (mm)	Thickness (in)	Quantity
Part of kit 4880	1.1	0.04	1
Part of kit 4880	1.3	0.05	1
Part of kit 4880	1.5	0.06	1



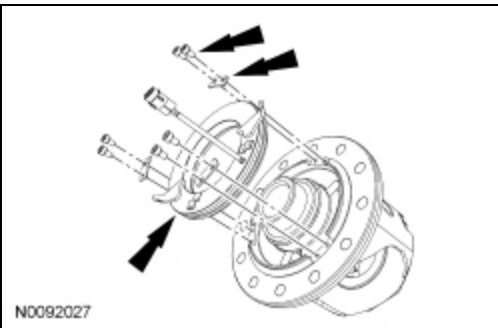
10. Tighten the differential pinion shaft lock bolt to 30 Nm (22 lb-ft).

11. Install the plunger assembly.

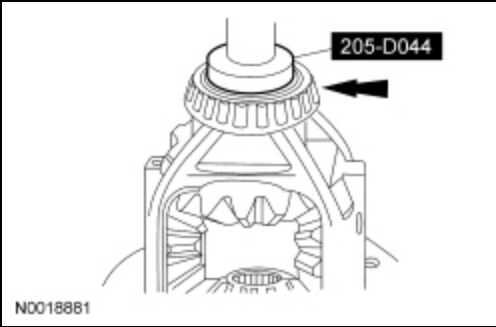


12. Install the ELD coil, 3 new stoppers and 6 new screws that hold the coil in place.

- Tighten to 5 Nm (44 lb-in).



13. Use the Differential Carrier Bearing Installer to install the 2 differential bearings.



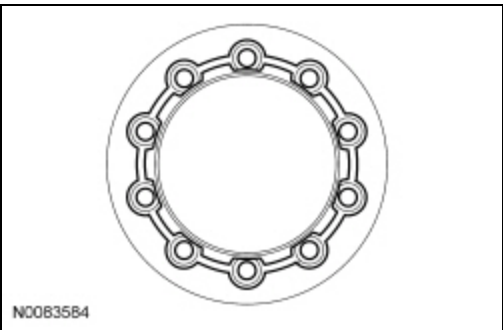
Electronic Locking Differential (ELD) equipped vehicles and all vehicles equipped with 3.55, 3.73, and 4.10 ratio gear sets

14. **NOTE:** *The differential flange and ring gear flange must be free of any old retaining compound material. Make sure both surfaces are clean and free of oil, dust and debris. Failure to clean the surfaces can result in ring gear runout concerns.*

Clean all traces of the old retaining compound material from the differential flange.

- Use solvent and Scotch-Brite® pads to remove.

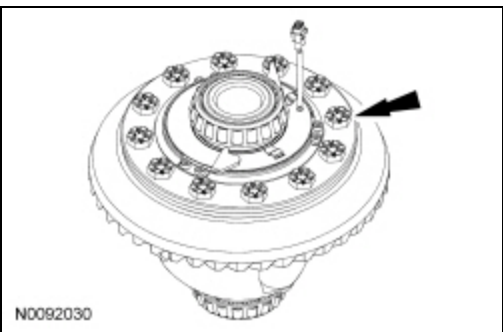
15. Apply a one-eighth inch bead of maximum strength retaining compound on the rear face of the ring gear in the pattern shown.



All vehicles

16. Install the ring gear and the 12 new differential ring gear bolts. Tighten in 2 stages.

- Stage 1: Tighten to 60 Nm (44 lb-ft).
- Stage 2: Tighten an additional 90 degrees.



17. **NOTE:** *Install the differential with the tabs on the ELD coil up against the differential bearing cap mounting surface.*

Install the differential carrier in the axle housing. For additional information, refer to Differential Carrier in this section.

